

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network
Security **COURSE CODE:** CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

M. Sadhana [192524386] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

M. Sadhana

Signature of the student:

Cryptography and Network

Reg no: 192524386
Name: M. Sadhana
Course code: CSA5116
Date: 06/10/2026

ASSIGNMENT-1

Application-based questions.

2). Scenario: Intercepted Suspicious Message.

Ciphertext $\rightarrow$ GSRH RH Z HVXJVG.

a). Most Likely original Message:

the most Likely original message

"THIS IS A SECRET"

The message is encrypted using the atbash substitution cipher, where each letter is replaced by its opposite.

	a	b	c	d	e	f	g	h	i	j	k	l	m	n
	0	1	2	3	4	5	6	7	8	9	10	11	12	13
	o	b	q	r	s	t	u	v	w	x	y	z		
	14	15	16	17	18	19	20	21	22	23	24	25		

Cipher	G	S	R	H	R	H	Z	H	V	X	J	V	G
Plain	T	H	I	S	I	S	A	S	E	C	R	E	T

Therefore:

GSRH RH Z HVXJVG $\rightarrow$ THIS IS A SECRET.

Step: 1

Separate the words

GSRH RH Z HVXJVG

Step: 2

First word: GSRH | second word: RH | Third word:

G → T

S → H

R → I

H → S

R → I

H → S

A — Z

Fourth word: HVXJVG

H → S

V → E

X → C

J → R

G → T

Therefore,

HVXJVG → SECRET

Step: 3

Combine all decoded words.

GSRH RH Z HVXJVG

↓ ↓ ↓ ↓
THIS IS A SECRET

original message:

"THIS IS A SECRET"

b). Justification:

the answer can be justified using the following observations.

1). Repeating patterns.

The ciphertext has spaces and word lengths:

GSRH = 4 letters

RH = 2 letters

Z = 1 letter

HVXIVG = 6 letters

This matches the common English sentence patterns.

THIS = 4 letters

IS = 2 letters

A = 1 letter

SECRET = 6 letters

So, the structure strongly suggests,

2). Simple substitution pattern.

Each ciphertext letter consistently represents a cipher letter.

for eg,

G	→	T
S	→	H
R	→	I
H	→	S

The same substitution rule works throughout the message.

3). Atbash characteristic

The alphabet is reversed:

A $\leftrightarrow$ Z

B $\leftrightarrow$ Y

C $\leftrightarrow$ X

D $\leftrightarrow$ W

E $\leftrightarrow$ V

....

G $\leftrightarrow$ T

H $\leftrightarrow$ S

I $\leftrightarrow$ R

R $\leftrightarrow$ I

S $\leftrightarrow$ H

V $\leftrightarrow$ E

X $\leftrightarrow$ C

Applying the rule gives.

GSRH RH Z HVXIVG $\rightarrow$ THIS IS A SECRET

Conclusion:

therefore, the intercepted message is most likely

"THIS IS SECRET"

3). Scenario: Secure Messaging System.

plain text : HELLO

KEY word : KEY

the encryption technique is the Vigenere cipher.

Demonstrate the Encryption process step by step.

Step: 1

write the plaintext.

HELLO

There are 5 characters.

Step: 2

write the keyword.

the key word is,

KEY

the plaintext has 5 letters, but the keyword has only 3 letters.

Step: 3

convert letters into numbers

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12
n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

Step: 4

Encryption

$$C = (p + k) \bmod 26$$

$$H = (7 + 10) \bmod 26 = 17 = R$$

	Plain Text	Value	Key	Key Value
1	H	7	K	10
2	E	4	E	4
3	L	11	Y	24
4	L	11	K	10
5	O	14	E	4

$$E = (4 + 4) \bmod 26 = 8 \Rightarrow I$$

$$L = (11 + 24) \bmod 26 = 35 \Rightarrow 9 \Rightarrow J$$

$$L = (11 + 10) \bmod 26 = 21 \Rightarrow V$$

$$O = (14 + 4) \bmod 26 = 18 \Rightarrow S$$

$$\begin{array}{r} 1 \\ 26 \overline{) 35} \\ \underline{26} \\ 09 \end{array}$$

Step: 5

Write the final ciphertext

plain text $\rightarrow$ HELLO

key word $\rightarrow$ KEYICE

ciphertext $\Rightarrow$ RJJSVS

Encryption:

plain text + key = ciphertext

Decryption:

ciphertext - key = plaintext

$$C = (P + K) \bmod 26$$

$$C = (P - K) \bmod 26$$